What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

An ideal asteroid orbit has nearest distance **3 AU** and farthest distance **6 AU**. Where is it fastest? Explain with the equal-area law.

Word help: orbit = a path around another body under gravity

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A **2 kg** cart and a **4 kg** cart move right at **3 m/s**. Their momenta mv are **6** and **12 kg·m/s**. Stopping the second requires twice the momentum change.

Your turn: 1. **Draw the velocity** and momentum arrows for both carts.
2. **Explain which pair has equal** lengths.

Sentence starter: I think … . I can point to …, which shows …, so … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Carts A and B both move right at **2 m/s**. A has mass **1 kg**; B has mass **3 kg**. Which has more momentum? **State the controlled quantity**.

Sentence starter: I notice … . This supports … because … .



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

In the track test from question **3**, cart A has mass **1 kg** and cart B has mass **3 kg**. Both initially move right at **2 m/s** and stop in **3 s**. **Complete the table**, then **calculate each average stopping force**. **State the direction**.

Sentence starter: I found … [unit]. This value means … in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage? Explain **0**.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Use the supplied gravity values: Gm_1m_2=**144 N·m²** and separation **6 m**. **Find the force** and its direction. **Label the masses**, field or force relevant to the question.

Word help: separation = distance between the specified objects or centers

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** cart travels right at **3 m/s** and stops in **3 s**.
1. **Find the magnitude of its** momentum change and average stopping force.
2. **Draw the stopping-force arrow**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Cart stopping

Right is positive.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find a lighter**, faster cart with the same momentum as a **4 kg** cart at **3 m/s**.
2. **Explain why equal momentum** does not mean equal mass or speed.

Sentence starter: I think … . I can point to …, which shows …, so … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3